Task 1: Build the Library schema. Create library_db with these tables, using the right data types:

- member: id (auto-generated), full name (required), email (required, unique), joined date (defaults to today), status (ACTIVE or SUSPENDED, default ACTIVE).
- book: id, ISBN (13 characters, unique), title (required), price (must not be negative), copies (default 1, must not be negative).
- loan: id, member, book, issued date (default today), due date (must not be before issued), returned date (optional). Foreign keys to both parents, and the same member cannot borrow the same book on the same day.

Task 2: Prove the constraints work. Insert two valid members and two valid books. Then run each statement below and record the error number.

Attempt	**Expected error**
Loan for member 999	1452 (foreign key)
Book with price -5	3819 (CHECK)
Two members with the same email	1062 (duplicate)
Member with NULL email	1048 (NOT NULL)
Loan with due_on before issued_on	3819 (CHECK)

Task 3: Composite key. Create a book_author junction table for the many-to-many relationship between books and authors, with a composite primary key and two foreign keys. Insert 3 rows, then try to insert a duplicate pair and record the error.

Task 4: ER to tables. A hospital has doctors, patients, departments and appointments. A doctor belongs to one department. A patient can book many appointments. Draw the E-R diagram, create the tables with all keys and constraints, and insert 3 rows per table.

Setup

sql

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY, full_name VARCHAR(60) NOT NULL,
  department VARCHAR(30), salary DECIMAL(9,2), hire_date DATE);
INSERT INTO employee VALUES
```

(1,'Ayesha Rahman','Sales',45000,'2024-01-15'),

(2,'Rafiq Hasan','IT',62000,'2023-06-01'),

(3,'Nusrat Jahan','IT',58000,'2024-03-20'),

(4,'Imran Kabir','HR',NULL,'2022-11-05'),

(5,'Tania Akter','Sales',47000,'2025-02-10');

#	Task	Expected result
1	List IT employees earning over 55000, highest first.	Rafiq 62000, Nusrat 58000
2	List names starting with A or T.	Ayesha, Tania
3	Show all employees, newest hire first.	Tania, Nusrat, Ayesha, Rafiq, Imran
4	Show full_name and salary * 12 as annual_salary.	5 rows, Imran's value is NULL
5	Find employees with no salary, then display 0 instead of NULL.	Imran
6	Give Sales a 5% raise, checking with SELECT first.	Ayesha 47250, Tania 49350
7	Headcount and average salary per department.	Sales 2 / 46000, IT 2 / 60000, HR 1 / NULL
8	Keep only departments with 2 or more people.	Sales, IT
9	Label salary as High (60000+), Medium (50000+), Standard, or "Not set" when NULL.	Rafiq High, Nusrat Medium, Ayesha and Tania Standard, Imran Not set
10	Delete employees hired before 2023.	Imran removed, 4 rows remain

Task 11: Upsert. In one statement, change Nusrat's salary to 60000 and add a new employee (emp_id 6) using INSERT ... ON DUPLICATE KEY UPDATE. Afterwards the table has 6 rows and Nusrat earns 60000.

Setup

sql

```
CREATE TABLE sales (
  sale_id INT PRIMARY KEY, region VARCHAR(20), product VARCHAR(30),
  category VARCHAR(20), qty INT, unit_price DECIMAL(8,2), sale_date DATE);
INSERT INTO sales VALUES
 (1,'North','Laptop','Electronics',2,800.00,'2026-10-02'),
 (2,'North','Mouse','Accessories',10,12.50,'2026-10-05'),
 (3,'South','Laptop','Electronics',1,800.00,'2026-10-07'),
 (4,'South','Keyboard','Accessories',5,25.00,'2026-10-11'),
 (5,'East','Monitor','Electronics',3,150.00,'2026-10-15'),
 (6,'East','Mouse','Accessories',20,12.50,'2026-10-18'),
 (7,'West','Monitor','Electronics',2,150.00,'2026-10-21'),
 (8,'West','Keyboard','Accessories',NULL,25.00,'2026-10-25');
```

Revenue means qty * unit_price.

#	Task	Expected result
1	Compute COUNT(*), COUNT(qty), SUM(qty), AVG(qty).	8, 7, 43, 6.14
2	Revenue per region, highest first.	North 1725, South 925, East 700, West 300
3	Show only regions above 800.	North, South
4	Revenue per category.	Electronics 3150, Accessories 500
5	Regions above 400 revenue, for sales from 2026-10-10 onward.	East (700)
6	Label each sale Large (800+), Medium (200+), Small, or “No quantity”.	Laptop 1600 and 800 Large, Keyboard (NULL) “No quantity”

#	Task	Expected result
7	One row per region with separate Electronics and Accessories revenue.	North 1600 / 125, South 800 / 125, East 450 / 250, West 300 / NULL
8	Sales whose revenue is above the average sale (scalar subquery).	Both Laptop sales (average is 521.43)
9	Regions that sold a product priced over 500 (IN).	North, South
10	Total revenue per product, using a derived table or CTE.	Laptop 2400, Monitor 750, Mouse 375, Keyboard 125
11	With a CTE and CASE, tier each region: Top (1000+), Mid (500+), Low.	North Top, South Mid, East Mid, West Low
12	With two chained CTEs, find regions above the average regional revenue.	North, South (average is 912.50)

Task 13: Reporting problem. Build one summary row per region with sales count, units, revenue, percentage of total revenue (1 decimal place), and a status of “Target met” at 800 or more, otherwise “Below target”. Sort by revenue, highest first.

Expected: North 47.3% and South 25.3% are “Target met”; East 19.2% and West 8.2% are “Below target”.

Task 14: Break it on purpose. Run each of these, read the error, and fix it.

- SELECT region, product, SUM(qty) FROM sales GROUP BY region; (error 1055)
- SELECT region FROM sales WHERE SUM(qty) > 10 GROUP BY region; (aggregate in WHERE)
- A derived table with no alias.